Shouzhi Fang

Ph.D. Student, Computer Science and Engineering, Syracuse University

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http://frank1412.github.io Google Scholar

Research Interests

Computer architecture and processing in memory; Hardware and systems security, including isolation in memory systems, GPUs and accelerators; Encryption accelerators and trusted execution environments; Program analysis and automated testing.

Education

2024 – present **Ph.D. Computer Science and Engineering, Syracuse University** in NY, USA.
Co-advised by Prof. Alex K. Jones and Prof. Endadul Hoque.

2021 – 2023 **M.Sc. Computer Science, Rutgers University** in NJ, USA.

2015 – 2019 **B.E. Computer Science and Technology, Chongqing University of Posts and Telecommunications** in Chongqing, China.

Research Publications

Conference Proceedings

1. **S. Fang**, W. C. Tegge, C. Yue, E. Hoque, and A. K. Jones, "Memory is the cipher: Confidential AES in commodity DRAM," in *Proceedings of the Great Lakes Symposium on VLSI 2026*, ser. GLSVLSI '26, Canandaigua, NY, USA: ACM, Jun. 2026, pp. 317–324. DOI: 10.1145/3787109.3815271.

Preprints

1. **S. Fang**, W. C. Tegge, M. O. Faruque, P. Zhou, E. Hoque, and A. K. Jones, "YAVIN: A unified architecture for secure edge processing in memory," *arXiv preprint arXiv:2608.13496*, 2026. DOI: 10.48550/arXiv.2608.13496. arXiv: 2608.13496 [cs.AR].
2. W. C. Tegge, J. P. C. de Lima, **S. Fang**, J. Castrillon, and A. K. Jones, "RAPID: Row-parallel arithmetic processing in DRAM," *arXiv preprint arXiv:2610.02502*, Oct. 2026. DOI: 10.48550/arXiv.2610.02502. arXiv: 2610.02502 [cs.AR].

Posters

1. **S. Fang**, W. C. Tegge, C. Yue, A. K. Jones, and E. Hoque, *Memory is the cipher: Confidential AES in commodity DRAM*, Research Day at Syracuse University, Syracuse, NY, USA, Mar. 2026.

Experience

2024 – present **Syracuse University**, Research Assistant, Syracuse, NY.
Research on processing-in-memory (PIM) architectures and memory-system security: confidential AES computed inside commodity DRAM (first author, GLSVLSI'26); YAVIN, a unified architecture for secure edge PIM (first author); and RAPID, row-parallel arithmetic processing in DRAM (co-author).

Experience (continued)

2023 – 2024 **Black Cow Tech**, Software Engineering Intern, Remote, USA.
• Built an end-to-end CI/CD pipeline for an audio transcription service on AWS (IAM, S3 event triggers, Lambda; React and Spring Boot), with automated code-quality checks, unit tests and deployment.
• Automated builds, testing and releases with GitHub Actions, added automatic rollback for failed deployments, and managed infrastructure changes through AWS CloudFormation.

2020 – 2021 **Business-intelligence of Oriental Nations Corporation**, Backend Software Engineer, Beijing, China.
• Built a document-content extraction pipeline around the EAST text detector (Python, PyTorch), served behind Nginx load balancing at up to 30,000 concurrent requests.
• Developed a distributed sentiment-analysis system on Hadoop (Java, HDFS) and Spark (PySpark) for near-real-time large-scale data processing.

2020 **Jiangsu Phoenix Union Education Technology Co.**, Full-Stack Software Engineering Intern, Nanjing, China.
• Improved throughput of an epidemic-prevention system with Redis caching and RabbitMQ for reliable message delivery; built RESTful APIs, database schema and a Vue.js front end.
• Developed a Spring Boot and MyBatis education platform, including a Redis-backed flash-sale module that prevents overselling under bursts of concurrent orders.

Professional Service

Journals Reviewer for IEEE Transactions on Computers.

Conferences Reviewer for ISCA 2026, MICRO 2026, HPCA 2026.

Miscellaneous Experience

Awards and Achievements

2019 **Outstanding Graduate** awarded by the Computer Science and Technology Department at CQUPT.

2016–2018 **Second Prize Scholarship**, awarded by CQUPT.

Security Vulnerability Disclosures

2026 **NVIDIA Multi-Instance GPU (MIG)**: Discovered and reported a cross-instance fault-isolation issue in which a workload in one GPU instance can crash workloads in a co-located instance. Reported to NVIDIA PSIRT; coordinated disclosure in progress.

Other

2007 **Go (Weiqi), 5-dan amateur**, certified by the Chinese Weiqi Association.